

Erect and Tape a Carton SOP

One episode builds one carton. A flat blank is pushed square into a tube, the four bottom flaps are folded in, three strips of tape seal the bottom in an H pattern, and the carton is turned over onto the packing spot with its open end facing up. The episode ends with an empty cradle and a bare peel board.

The order is always the same: square the tube, fold the short flaps, fold the long flaps, lay the seam strip, lay the first end strip, lay the second end strip, turn the carton over. Only fold flaps on a tube that is already square, and only lay tape on a seam that is already closed.

Every strip goes down the same way, in three moves: the left gripper holds the carton still, the right gripper lays the strip onto the seam, and the right gripper presses it down. The left arm never lays tape and the right arm never holds.

Never reach across to the far end of the carton. Every fold, every lay, and every press happens at the carton's **right end** or along its **near side**. When the work is at the other end, turn the cradle a half turn so that end comes around to the right. The carton comes to the arm; the arm does not stretch out to the carton.

The carton stays in the cradle until Step 7. While it is in the air it is never rotated more than a quarter turn at a time, and a gripper only ever opens over the carton itself or over the packing spot.

The table is set up in one of three ways. Only the peel board, with its three strips, moves; the cradle, the blank, and the packing spot are in the same place in all three.

- **Config M:** the peel board is at the front-center, in front of the cradle.
- **Config R1:** the peel board is on the right, beside the cradle.
- **Config R2:** the peel board is in the back-right corner.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the peel board stands at the front-center (**Config M**), the right (**Config R1**) or the back-right (**Config R2**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the **right gripper** fetches every strip from the peel board in every config. The board never stands on the left, because the left gripper is pinned on the carton whenever a strip is fetched and the left arm never touches a strip. No arm reaches across the table.

- **Fixed roles:** everything else is the same in all three configs — the left arm holds, the right arm folds, turns, lays, and presses, and both grippers turn the carton over in Step 7.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the cradle in the middle, the peel board in the place for this episode's config, and the packing spot on the near right.
3. The top of the carton is fully in frame from above — both seating lines, all four flaps, the center seam from end to end, and both end seams.
4. The near face and both end panels of the carton are visible, so every tape overhang can be seen being pressed down.
5. Both arms are at home with their grippers open.
6. The tabletop is clear of everything except the zones listed below.
7. The right gripper can reach the **left end of the center seam** without the arm leaning out over the carton. If it cannot, move the cradle to the right until it can.
8. The right gripper can reach the **free edge of the far long flap** without the arm going to the end of its reach. If it cannot, move the cradle toward the front edge until it can.
9. The two arms clear each other along the seam line — test this with the right gripper on the seam and the left gripper on the near long flap beside it.
10. Nothing blocks the paths between the peel board, the cradle, and the packing spot, and the right gripper reaches every grab tab on the board — at the front-center (Config M), the right (Config R1), or the back-right (Config R2) — without going to the end of its reach.

Materials checklist

1. One flat carton blank stands on edge in the cradle channel, leaning back against the far channel wall, with its **left spine on the left seating line**.
2. The blank is a plain four-panel tube with four flaps at each end. It is uncrushed, and all four corner creases open to a right angle easily by hand.
3. The flap end pointing **up** is the bottom of the carton. Both ends of the blank are identical, so nothing needs marking: whichever end faces up is the end that gets taped.

4. The flap end resting on the channel floor is left alone for the whole episode. Its four flaps hang in line with their panels, and the tube stands on their edges.
5. Three tape strips lie on the peel board, sticky side down. Each one peels off cleanly in a single piece and has a stiff folded-back **grab tab** at one end. The board stands in one place for the whole episode:
 - **Config M:** front-center, in front of the cradle, clear of the turn tab's swing
 - **Config R1:** right, beside the cradle
 - **Config R2:** back-right corner
6. One **seam strip** sits on the seam strip spot. It is longer than the carton's long side and lies left to right, with its grab tab at the **right** end.
7. Two **end strips** sit on the end strip spots. Each is longer than the carton's short side and lies near to far, with its grab tab at the **near** end (toward the front edge).
8. Each strip is long enough to hang past both ends of its seam and fold down onto the panel below, but not so long that the overhang runs past the bottom of that panel.
9. The cradle is at its **home stop**, its channel holds nothing but the blank, and both seating lines are visible.
10. The packing spot is bare.

Workspace layout

- **Cradle:** in the middle of the tabletop, centered between the two arms, on a turntable base. It is a channel with a **near wall** and a **far wall** set one carton width apart, and no end walls, so a flat blank can stand in it and overhang both ends. Both walls are no taller than half the carton, so every tape overhang can still be pressed down below the strip. The carton stays in the cradle from Step 1 through Step 6.
- **Seating lines:** two lines printed across the channel floor, one near each end. When the carton is square and centered, one end panel sits on each line. These lines are what "centered" means here, and they look the same after the cradle turns a half turn.
- **Turn tabs:** two tabs on the turntable base, one at its **near right** corner and one at its **far left** corner. A half turn brings the far left tab around to the near right corner, so there is always a tab where the right gripper can grab it. The base has two hard **turn stops**, half a turn apart.
- **Peel board:** at the front-center in front of the cradle (**Config M**), on the right beside the cradle (**Config R1**), or in the back-right corner (**Config R2**), always within reach of the right arm and clear of the turn tab's swing. It holds the three strips in the order they are used, with the closest to the cradle used first: the seam strip, then the two end strips in a row further out. Its surface releases a strip cleanly when the grab tab is lifted.
- **Packing spot:** on the near right of the tabletop, kept bare. The finished carton is set here, and nothing else is ever put on it.

- **Left side of the tabletop:** kept clear. The left arm only ever works on the carton itself, approaching it from the left. Nothing is staged or set down on that side.

Vocabulary

- **Arm assignment:** the left arm holds. It braces the blank while it is squared, and it holds the carton still for every fold and every strip. The right arm does all the work: every push, fold, turn, lay, and press, and it fetches every strip from the peel board wherever the board stands — front-center (Config M), right (Config R1), or back-right (Config R2). Neither arm takes over the other's job. The only move where they share the work is the turn over in Step 7, where both grippers hold the carton at the same time.
- **Reach rule:** the left arm only touches the carton's **left end** and its **middle**. The right arm only touches the carton's **right end** and its **near side**. The one motion that crosses the full width of the carton is the seam press stroke in Step 4, which runs left to right at a single constant depth and still does not lean out over the carton.
- **Blank:** the carton while it is still flat and folded shut, before it is squared.
- **Spine:** one of the two folded edges of the flat blank — one at its left end, one at its right end. Each spine becomes a corner of the squared carton.
- **Tube:** the blank once it is square — four panels, four right-angle corners, open at both ends.
- **Squared:** the near and far panels lie flat against the channel walls with no bowing, all four corners are open to right angles, and an end panel sits on each seating line.
- **Bottom end:** the flap end facing up. It is the only end that gets folded and taped.
- **Top end:** the flap end resting on the channel floor. Nothing is done to it, and after Step 7 it is the end facing up, ready for packing.
- **Long flaps:** the two wide bottom flaps, on the carton's **near** and **far** sides. They meet each other at the center seam.
- **Short flaps:** the two narrow bottom flaps, at the carton's **left** and **right** ends. They fold in first and end up trapped underneath the long flaps.
- **Center seam:** the line where the two long flaps meet, running left to right down the middle of the carton's bottom.
- **End seam:** the line where the folded long flaps meet an end panel, running from near to far at each end of the carton. There are two of them.
- **Fold:** the right gripper pinches a flap in the middle of its free edge, swings it in until it lies flat and level, holds it there for 2 seconds, then releases.
- **Brace:** the closed left gripper presses against a spine and holds it still while the right gripper pushes the other spine toward it. A brace stays in one place — it never slides along.
- **Pin:** the closed left gripper presses firmly on the carton to hold it still — straight down on a flap or a seam, or in against a panel. A pin never pinches and never slides, and it is held for the whole step.

- **Half turn:** the right gripper pinches the **near right turn tab** and swings it along the near edge to the **left** in one continuous motion, until the base meets its stop. Every half turn goes the same direction, every time. Both grippers are clear of the carton while it turns.
- **Start zone:** where the peel board and its three strips stand for the episode — front-center (**Config M**), right (**Config R1**), or back-right (**Config R2**). One per episode, chosen before recording and never changed mid-episode.
- **Strip:** one length of tape, already cut to size. It is only ever handled by its **grab tab**.
- **Grab tab:** the folded-back end of a strip, which has no sticky side exposed. It is the only part of a strip a gripper ever touches.
- **Overhang:** the part of a strip that hangs past the end of its seam and gets pressed down onto the panel below.
- **Hangs straight:** the strip dangles free and flat from the lifted grab tab, without curling, twisting, or sticking to itself. A strip that is not hanging straight is thrown away — do not try to straighten it.
- **Lay:** the right gripper lowers the strip's free end onto the start of a seam, then travels along the seam so the strip settles onto it behind the gripper, and releases the tab at the far end.
- **Press stroke:** the closed right gripper presses down on a laid strip and draws along it in one continuous stroke from end to end, without lifting off.
- **Quarter turn:** both grippers press on the carton's two end panels, lift it straight up clear of whatever it is standing on, roll it toward the front edge through a quarter turn about the line between the two grippers, and lower it until it rests. The carton is turned over using two quarter turns with a set-down in between — never one continuous half turn.
- **Settled:** the carton or flap stays put for 2 seconds after the gripper lifts away, with nothing springing back, rocking, or sliding.
- **Release point:** the carton, or the packing spot. A gripper never opens anywhere else — in particular, never over the peel board or over the bare tabletop.

Steps

Steps 1 through 7 cover one carton. Every fold is taken in the middle of the flap's free edge and held for 2 seconds before the gripper opens, and every strip is carried in the same orientation it will be laid in.

Only the strip fetches in Steps 4, 5, and 6 depend on the config: the **right gripper** reaches to the peel board at the front-center (Config M), the right (Config R1), or the back-right (Config R2). Every other line is the same in all three configs.

Step 1: Square the blank into a tube

Goal: the blank stands in the channel as a square tube, with an end panel on each seating line.

- The **left gripper**, closed, braces the **left spine** on the **left seating line**, pressing it from the left. It holds there for the whole step.
- The **right gripper**, closed, presses on the **right spine** and pushes it straight along the channel toward the left spine in one continuous motion, until the near and far panels lie flat against the channel walls and the **right end panel sits on the right seating line**.
- Both grippers lift clear.

Push along the channel, not across it. A push angled into the far wall folds the blank flatter instead of opening it up, and the corner creases set the wrong way.

Check: all four corners are open to right angles, the near and far panels lie flat against the channel walls with no bowing, both end panels sit on their seating lines, all four bottom flaps stand up in line with their panels, and the tube holds its shape for 2 seconds after both grippers leave. If a corner is still creased flat, a panel bows, or the tube springs back, brace the left spine again and push the right spine the rest of the way.

Step 2: Fold the two short flaps in

Goal: both short flaps folded flat and level across the bottom end, one at each end of the carton.

- The **left gripper**, closed, presses on the **outer face of the near panel** at the middle of the carton, holding it in against the far channel wall, and holds it there.
- The **right gripper** pinches the **right short flap** in the middle of its free edge, swings it in over the open end until it lies flat and level, holds it for 2 seconds, and releases.
- Both grippers lift clear of the carton. The **right gripper** pinches the **near right turn tab** and swings it left in one motion until the base meets its stop, then releases the tab and lifts clear.
- The **left gripper** presses on the near panel at the middle again. The **right gripper** folds the short flap that is now at the right end, the same way.
- Both grippers lift clear.

The cradle turns; the arm stays where it is. Reaching across the carton for the far flap is what pulls the tube out of square and what forces the arm to stretch.

Check: both short flaps lie flat and level across the bottom end — neither one standing up, curling, or overlapping the other — and the tube is still square with both end panels on their seating lines. If a flap has sprung back up, fold it again and hold it longer before releasing.

Step 3: Fold the two long flaps down

Goal: both long flaps down and meeting edge to edge along the center seam, with the short flaps trapped flat underneath.

- The **left gripper**, closed, pins the **left short flap** flat and holds it there for the whole step.

- The **right gripper** pinches the **far long flap** in the middle of its free edge, swings it toward the front edge until it lies flat and level over both short flaps, holds it for 2 seconds, and releases.
- The **right gripper** pinches the **near long flap** in the middle of its free edge, swings it away from the front edge until its free edge meets the far flap's free edge along the middle of the carton, holds it for 2 seconds, and releases.
- Both grippers lift clear.

The far flap goes down first. If the near flap is already down, the gripper has to reach out over it to get to the far one, which drags the near flap out of line and puts the arm exactly where it should never be.

Check: both long flaps lie flat and level, their free edges meet along the middle of the carton in one straight seam with no overlap and no gap wider than the seam strip can cover, both short flaps are trapped flat underneath with no corner poking out, and nothing stands up above the flap line. If the flaps overlap or a corner stands up, fold the near flap back up and lay it down again.

Step 4: Lay and press the seam strip

Goal: the seam strip runs down the whole center seam, pressed flat, with its right overhang folded down.

- The **left gripper**, closed, pins the **near long flap at its left end**, clear of the seam, and holds it there for the whole step.
- **IF the peel board is at the front-center (Config M):** the **right gripper** reaches forward, pinches the **grab tab** of the **seam strip**, lifts it straight up off the peel board until it hangs straight and clear, and carries it back over the near wall to the carton without rotating it.
- **IF the peel board is on the right (Config R1):** the **right gripper** pinches the **grab tab** of the **seam strip**, lifts it straight up off the peel board until it hangs straight and clear, and carries it left to the carton without rotating it.
- **IF the peel board is at the back-right (Config R2):** the **right gripper** reaches back, pinches the **grab tab** of the **seam strip**, lifts it straight up off the peel board until it hangs straight and clear, and carries it forward and left to the carton without rotating it.

Then, in all three:

- Lower the strip's free end onto the seam at the carton's **left end**, so the overhang sticks out past the left end panel. Travel right along the seam, letting the strip settle onto it behind the gripper, until the tab end sits past the right end panel. Release the tab.
- The **right gripper**, closed, presses down on the strip at the left overhang and draws along the seam in one continuous stroke, left to right, all the way to the right overhang, without lifting off.

- The **right gripper** presses the **right overhang** down onto the right end panel.
- Both grippers lift clear.

Leave the left overhang alone for now. It gets pressed in Step 6, once the half turn has brought it around to the right. Do not reach for it.

Check: the strip runs down the middle of the seam from end to end, covering it along its whole length, lying flat with no bubbles, folds, or wrinkles, and the right overhang lies flat against the end panel. If the strip landed off the seam, folded on itself, or lifted at an edge, peel it off, throw it away, and lay a fresh seam strip. Never peel a strip up and stick it back down.

Step 5: Lay and press the first end strip

Goal: an end strip across the right end seam, pressed flat, with both overhangs folded down.

- The **left gripper**, closed, pins the **center seam at the middle of the carton** and holds it there for the whole step.
- **IF Config M:** the **right gripper** reaches forward to the peel board. **IF Config R1:** it reaches right. **IF Config R2:** it reaches back to the back-right corner.
- The **right gripper** pinches the **grab tab** of the first **end strip**, lifts it straight up off the peel board until it hangs straight and clear, and carries it to the carton without rotating it.
- Lower the strip's free end onto the **right end seam** at the far side, so the overhang sticks out past the far panel. Draw toward the front edge, letting the strip settle onto the seam behind the gripper, until the tab end sits past the near panel. Release the tab.
- The **right gripper**, closed, presses down on the strip at the far overhang and draws along the end seam in one continuous stroke toward the front edge, all the way to the near overhang, without lifting off.
- The **right gripper** presses the **far overhang** down onto the far panel, then the **near overhang** down onto the near panel.
- Both grippers lift clear.

Check: the strip covers the right end seam from end to end, crossing over the seam strip in the middle, lying flat with no bubbles, folds, or wrinkles, and both overhangs lie flat against their panels. If the strip landed off the seam or folded on itself, peel it off, throw it away, and use the second end strip in its place — then take a fresh strip for Step 6.

Step 6: Turn the cradle and finish the second end

Goal: the H is complete — the second end strip is down and every overhang is pressed.

- With both grippers clear of the carton, the **right gripper** pinches the **near right turn tab** and swings it left in one motion until the base meets its stop, then releases the tab and lifts clear.
- The **right gripper** presses down the seam strip's remaining overhang, which is now at the right end, onto its end panel.

- The **left gripper**, closed, pins the **center seam at the middle of the carton** and holds it there.
- **IF Config M:** the **right gripper** reaches forward to the peel board. **IF Config R1:** it reaches right. **IF Config R2:** it reaches back to the back-right corner.
- The **right gripper** lays and presses the second **end strip** across the end seam that is now on the right, exactly the same way as in Step 5: let it hang straight, lower the free end at the far side, draw toward the front edge, release the tab, press from far to near in one stroke, then press the far overhang and the near overhang down.
- Both grippers lift clear.

The second end uses the same motion as the first, not a mirrored version of it. That is the whole point of turning the cradle.

Check: the bottom shows one strip down the center seam and one strip across each end seam, forming an H; all overhangs from the seam strip and both end strips lie flat against their panels; no seam line shows through a gap between strips; and the carton is still square with both end panels on their seating lines. If an overhang has lifted, press it down again with the closed gripper.

Step 7: Turn the carton over onto the packing spot

Goal: the carton stands on its taped bottom at the packing spot, open end up.

- The **left gripper**, closed, presses on the **left end panel** and the **right gripper**, closed, presses on the **right end panel**, both at the middle of the panel's height.
- Both grippers lift the carton straight up until it clears the channel walls.
- Both grippers roll the carton **toward the front edge** through one **quarter turn**, keeping hold the whole time, until the taped bottom faces the front edge. Lower it onto the packing spot so it comes to rest on its near panel. Release both grippers and lift both clear.
- Both grippers press on the **left end panel** and the **right end panel** again and lift the carton straight up clear of the table.
- Both grippers roll it **toward the front edge** through a second **quarter turn**, until the taped bottom faces down, and lower it straight down onto the packing spot until it rests. Release both grippers and lift both clear.

Set the carton down between the two turns. Doing both turns in one continuous roll runs both wrists to their limit, and an arm that stalls partway around will drop the carton or drag it across the spot.

Check: the carton stands square on its taped bottom at the packing spot, open end up, all four top flaps standing clear of the opening, no strip lifted or torn by the handling, and it is settled. If it lands on an edge or off the spot, take both end panels again, lift it clear, and lower it again.

Step 8: End the episode

- Confirm the carton stands taped-side down on the packing spot, the cradle channel is empty and back at its home stop, and the peel board is bare.
- Return both arms home, then stop recording.

Two half turns per episode bring the cradle back to the stop it started from. If it is not there, a turn was either skipped or run past its stop.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the taped carton off the packing spot and put it on the finished stack, off the tabletop. A taped carton is never reused.
2. Confirm the cradle sits at its home stop. If it does not, turn it back to the stop by hand.
3. Clear the channel floor and the tabletop of tape scraps, backing, and any discarded strip.
4. Stand a fresh flat blank on edge in the channel, leaning back against the far wall, left spine on the left seating line, with the end that will be taped facing up.
5. Reload the peel board: one seam strip on the seam strip spot, lying left to right with its grab tab at the right, and two end strips on the end strip spots, lying near to far with their grab tabs toward the front edge. Stand the board in the place for the next episode's config — front-center (Config M), right (Config R1), or back-right (Config R2).
6. Check that every grab tab is still folded back and free of tack, and that each strip lifts off the board in one piece.
7. Check the cradle: both channel walls firm, both seating lines visible, both turn tabs attached, and both turn stops holding the base square.
8. Replace anything damaged — a blank that is crushed, torn, or too stiff to square, a blank whose long flaps will not meet at the seam, a strip that has lost its tack, or a peel board that no longer releases a strip cleanly.
9. Wipe the tabletop and confirm the cradle and the peel board are firm and have not shifted.
10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R1 (peel board on the right, beside the cradle). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the peel board is not in the start zone for the config; a gripper reaches across the table for a strip; the left gripper leaves the carton to fetch a strip; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the right gripper fetches every strip from the peel board wherever it stands; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the peel board is before the first reach, then follow that config's IF lines through Steps 4, 5, and 6.

Violation: Wrong order

- **Visible cue:** a flap is folded before the tube is square, a long flap goes down before both short flaps are in, the near long flap goes down before the far one, a strip is laid before the flaps are down, or an end strip is laid before the seam strip.
- **SOP rule broken:** Steps 1 to 6 — the order never changes, and tape only goes on a seam that is already closed.
- **Coaching note:** square, short, long, seam, end, end. Nothing jumps the queue.

Violation: Arms swapped roles

- **Visible cue:** the left gripper folds a flap, turns the cradle, lifts a strip, or makes a press stroke; the right gripper braces or pins while the left one works; or the left gripper travels to the carton's right end.
- **SOP rule broken:** Steps 1 to 7 — the left arm holds and the right arm works, and no strip ever touches the left gripper.
- **Coaching note:** left holds, right works. Both arms move together in Step 7 and nowhere else.

Violation: Reached across instead of turning the cradle

- **Visible cue:** the right gripper works on the carton's **left end** — folding the left short flap, laying or pressing the left end strip, or pressing the left overhang — instead of turning the

cradle first; or the arm leans out over the carton to reach the far side of a flap or a seam.

- **SOP rule broken:** Steps 2, 5, and 6 — when the work is at the other end, turn the cradle a half turn so the work comes to the arm.
- **Coaching note:** if you are stretching, stop and turn the cradle. The turn is cheaper than the reach, and it is what keeps the carton square.

Violation: Half turn not run to the stop

- **Visible cue:** the base is left partly turned, rocked back and forth, turned the wrong way along the near edge, or turned more than once between the same two steps; or the episode ends with the cradle off its home stop.
- **SOP rule broken:** Steps 2, 6, and 8 — one continuous swing of the near right tab to the left, all the way to the stop, the same way every time.
- **Coaching note:** one tab, one direction, run it to the stop, and look at the stop.

Violation: Cradle turned by the carton

- **Visible cue:** a gripper pushes, twists, or drags the carton itself to turn the cradle instead of taking hold of the turn tab.
- **SOP rule broken:** Steps 2 and 6 — the turn is taken by the tab, with both grippers clear of the carton.
- **Coaching note:** the tab turns the cradle. The carton is just along for the ride.

Violation: Folded or taped without a hold

- **Visible cue:** a flap is folded or a strip is laid and pressed while the left gripper is off the carton; or the left gripper takes its hold only after the fold or the lay has already started, or comes off before the step is finished.
- **SOP rule broken:** Steps 2 to 6 — the hold is taken first and kept for the whole step.
- **Coaching note:** hold first, let go last.

Violation: Flaps folded on a tube that is not square

- **Visible cue:** a flap is folded while a corner is still creased flat, a panel is bowed off a channel wall, or an end panel is off its seating line.
- **SOP rule broken:** Steps 1 to 3 — the tube is squared before any flap is folded, and flaps are only ever folded on a square tube.
- **Coaching note:** the tube gets squared once, at the start. Everything after that is built on it.

Violation: Fold released before it held

- **Visible cue:** the gripper opens as soon as the flap is down and the flap springs back up, or the episode moves on with a flap standing, curled, or lifting at a corner.
- **SOP rule broken:** Steps 2 and 3 — hold the fold flat for 2 seconds before releasing.

- **Coaching note:** hold it down until it stops pushing back.

Violation: Strip taken by the wrong part

- **Visible cue:** a gripper pinches a strip anywhere other than its grab tab, touches the sticky face, or peels a strip off the board by dragging it instead of lifting the tab.
- **SOP rule broken:** Steps 4 to 6 — the grab tab is the only part of a strip a gripper touches.
- **Coaching note:** tab only. A strip that has touched the gripper is already ruined.

Violation: Strip laid without hanging straight

- **Visible cue:** the strip is curled, twisted, or stuck to itself as it is carried, and it gets laid anyway; or a gripper tries to unstick a folded strip in mid-air instead of throwing it away.
- **SOP rule broken:** Steps 4 to 6 — a strip that is not hanging straight is thrown away, not straightened.
- **Coaching note:** look at it hanging before you bring it over. A folded strip is scrap, not a repair job.

Violation: Strip laid off the seam

- **Visible cue:** a strip lands to one side of the seam it should cover, runs across it at an angle, starts short of the end, or leaves part of the seam line showing.
- **SOP rule broken:** Steps 4 to 6 — each strip covers its seam from end to end.
- **Coaching note:** put the free end on the seam before you start moving. The first end decides where the whole strip goes.

Violation: Strip peeled up and stuck down again

- **Visible cue:** a strip that is already laid gets lifted off and pressed back down, shifted along the seam after it has touched down, or repositioned by peeling one end.
- **SOP rule broken:** Steps 4 to 6 — a badly laid strip is peeled off, thrown away, and replaced with a fresh one.
- **Coaching note:** tape sticks once. Throw it away and take a new strip.

Violation: Press stroke skipped or broken

- **Visible cue:** the episode moves on from a laid strip with no press stroke; the gripper lifts partway along and sets down again; the strip is dabbed at in several places; or an end seam is pressed in the wrong direction.
- **SOP rule broken:** Steps 4 to 6 — one continuous press stroke, end to end, without lifting off.
- **Coaching note:** one stroke, all the way across, without coming off the tape.

Violation: Overhang left unpressed

- **Visible cue:** the end of a strip is left sticking out past the carton, hanging loose, or folded back on itself instead of pressed flat onto the panel below it.
- **SOP rule broken:** Steps 4, 5, and 6 — every overhang is pressed down onto its panel.
- **Coaching note:** the overhang is what holds the seal. Finish it before you move on.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the peel board, over the bare tabletop, or over open space, and a strip falls, drapes over the side of the carton, or lands anywhere other than on its seam.
- **SOP rule broken:** Steps 1 to 7 — a gripper opens over the carton or over the packing spot, and nowhere else.
- **Coaching note:** know where it will land before you open the gripper.

Violation: Strip dragged across the carton

- **Visible cue:** a strip is slid, dragged, or scrubbed along the flaps to position it, or it touches the carton somewhere other than the seam on its way down.
- **SOP rule broken:** Steps 4 to 6 — the free end is lowered onto the start of the seam, and the rest of the strip settles behind the gripper.
- **Coaching note:** lower it onto the seam, do not sweep it there.

Violation: Carton lifted or shifted in the cradle

- **Visible cue:** the carton is lifted, dragged, rotated, or pushed off its seating lines at any point before Step 7 — including being pulled out of the channel by a fold or a press stroke.
- **SOP rule broken:** Steps 1 to 6 — the carton stays in the cradle and stays on its seating lines until Step 7.
- **Coaching note:** the cradle is what keeps the carton square. Anything that moves it undoes Step 1.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides a flap, a strip, or the carton into position after releasing it, instead of making the correction the SOP calls for.
- **SOP rule broken:** Steps 1 to 7 — correct a fold by folding it again, a strip by throwing it away, and a placement by lifting it clear and lowering it again.
- **Coaching note:** do it again properly, or leave it alone.

Violation: Turn over done wrong

- **Visible cue:** the carton is turned with only one gripper, rolled away from the front edge, run through both quarter turns in one continuous roll with no set-down in between, dropped or tipped the rest of the way instead of lowered, or set down taped-side up or on its side.

- **SOP rule broken:** Step 7 — both grippers hold the end panels and roll the carton toward the front edge one quarter turn, set it down and release, then take it again for the second quarter turn.
- **Coaching note:** both hands, toward you, a quarter turn at a time, and let it rest in between.

Violation: Regripped in the air

- **Visible cue:** a gripper shifts, rolls, or takes a new hold on a strip or on the carton without setting it down first.
- **SOP rule broken:** Steps 4 to 7 — set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** a strip falls onto the tabletop or the floor, the carton is dropped, the cradle or the peel board is struck and shifts, a staged strip is knocked out of place, or the two arms hit each other along the seam line.
- **SOP rule broken:** Steps 1 to 7 — keep the travel path clear, and only release once the placement has settled.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a strip still on the peel board, an overhang unpressed, the carton still in the cradle or taped-side up, the cradle off its home stop, or an arm away from home.
- **SOP rule broken:** Step 8 — confirm the carton, the cradle, and the bare peel board, return both arms home, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, a controller-caused collision, or a motor error.
- **Turntable fault:** a stop that does not hold, a base that runs past its stop, or a turn tab that comes off in the gripper.
- **Defective blank:** crushed, torn, damp, creases too stiff to open to a right angle under a straight push, or a tube that springs back out of square once the grippers leave. Replace it before the next episode.

- **Blank cut wrong for the job:** long flaps that do not meet at the center seam, or flaps that overlap so far that the seam is not where the SOP says it is.
- **Defective strip:** lost its tack, stuck to the peel board, torn while being lifted, cut to the wrong length, or a grab tab that has come unfolded. Replace the strip before the next episode.
- **Peel board will not release** a strip in one piece.
- **Cradle or peel board shifts out of position** during the episode.

Annotation subtasks (from SOP)

1. Square a flat blank into a tube
2. Fold one flap in
3. Turn the cradle a half turn
4. Lay one tape strip onto a seam
5. Press one tape strip down and press its overhangs
6. Turn the carton a quarter turn and set it down
7. Return both arms home and end the episode